

ESSEN MULHEIM, Germany

WMO: 104100

Lat: 51.4042N

Lon: 6.9678E

Elev: 491

StdP: 14.44

Time Zone: 1.00 (EUC)

Period: 82-03

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	14.3	19.5	7.3	8.2	18.2	11.5	10.1	22.3	27.3	44.0	24.6	39.3	9.7	60	0.544

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.7	82.8	66.7	79.8	65.2	76.7	63.7	68.3	79.1	66.6	77.2	65.1	74.4	6.7	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.5	92.4	72.4	62.8	87.0	70.2	61.3	82.6	69.2	32.7	79.4	31.4	76.5	30.2	74.6	75.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 21.6	18.8	16.5	DB	14.4	88.5	9.7	3.0	7.4	90.7	1.7	92.5	-3.7	94.2	-10.8	96.3	(4)
(5)			WB	13.4	71.4	9.4	1.4	6.7	72.4	1.2	73.2	-4.1	74.0	-10.9	75.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	49.8	35.8	35.6	41.8	47.1	56.1	59.7	64.4	64.0	58.2	51.6	43.2	39.5		
	DBStd	12.20	8.63	8.53	6.44	6.75	7.09	6.65	6.28	5.64	5.59	5.95	6.66	6.26		
	HDD50	1868	439	404	264	134	21	3	0	0	2	52	219	330		
	HDD65	5720	904	822	719	537	290	189	92	89	216	416	654	792		
	CDD50	1809	0	2	11	47	209	293	446	433	248	102	15	3		
	CDD65	186	0	0	0	0	13	29	73	57	13	1	0	0		
	CDH74	1288	0	0	0	1	91	185	558	392	59	2	0	0		
	CDH80	259	0	0	0	0	8	23	135	88	5	0	0	0		
	WSAvg	8.1	10.2	9.6	9.3	8.3	7.2	6.9	6.8	6.5	6.9	7.6	8.6	9.3		
	PrecAvg	36.2	3.1	2.3	2.6	2.4	2.9	3.5	3.5	3.4	2.8	2.8	3.3	3.7		
(15) Precipitation	PrecMax	48.0	6.1	6.4	6.2	5.0	6.6	7.5	7.4	7.7	6.8	6.5	5.8	8.3		
	PrecMin	25.4	0.2	0.2	0.5	0.3	0.5	0.5	0.6	1.1	1.0	0.7	0.2	0.6		
	PrecStd	5.5	1.5	1.3	1.5	1.2	1.3	1.6	1.7	1.7	1.3	1.4	1.3	1.7		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	53.0	58.1	64.9	71.3	80.7	82.9	88.3	87.1	80.4	71.2	60.3	55.8		
		MCWB	49.6	50.5	51.5	56.5	62.3	67.5	68.4	67.0	65.1	61.4	51.6	50.9		
	2%	DB	50.0	53.2	58.5	65.8	76.6	79.2	83.6	82.2	74.8	66.2	57.1	53.5		
		MCWB	47.2	48.1	49.4	53.4	60.0	65.1	67.8	66.2	63.4	57.8	51.8	49.6		
	5%	DB	47.9	49.5	53.4	61.7	72.5	75.6	80.2	78.2	70.8	62.9	54.4	50.4		
		MCWB	45.5	45.5	47.2	51.1	58.5	62.9	65.7	64.7	61.7	56.5	50.4	48.1		
	10%	DB	46.1	46.3	50.3	58.4	68.8	71.8	76.5	74.7	67.1	59.6	52.4	47.9		
		MCWB	43.8	43.4	45.6	49.7	56.8	60.8	64.6	63.1	60.3	54.7	49.5	45.9		
	0.4%	WB	50.3	51.7	52.6	57.5	64.2	70.1	70.8	69.1	67.5	62.8	54.8	52.1		
		MCDB	52.0	55.9	62.0	68.2	78.3	79.4	82.9	80.9	76.5	69.7	57.4	54.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	48.0	48.9	50.5	54.4	61.4	66.8	68.9	67.0	64.7	59.5	53.0	49.8		
		MCDB	49.3	52.2	56.2	62.4	72.9	76.5	80.3	78.7	71.8	64.4	55.0	52.0		
	5%	WB	45.9	46.0	48.4	52.5	59.3	64.5	67.0	65.6	62.9	57.4	51.2	48.3		
		MCDB	47.3	48.9	52.5	60.0	70.4	73.6	77.7	76.1	69.0	61.4	53.6	50.5		
	10%	WB	43.8	43.6	46.5	50.2	57.5	62.2	65.3	64.2	61.1	55.5	49.6	46.2		
		MCDB	45.8	46.2	49.9	57.3	66.7	69.5	75.0	72.5	66.2	58.8	52.3	47.7		
	5% DB	MDBR	7.5	9.2	10.4	13.0	14.6	13.7	14.7	14.8	12.3	10.3	8.2	6.7		
		MCDBR	9.1	12.6	14.4	17.5	21.0	19.9	20.5	20.7	17.4	14.2	10.6	9.0		
		MCWBR	7.9	9.4	9.0	9.4	10.3	9.7	9.5	9.2	9.1	8.5	7.8	7.3		
		MCWBR	8.7	11.8	12.5	15.8	18.8	17.9	18.5	18.3	15.1	11.7	9.6	8.6		
(35) Mean Daily Temperature Range	5% WB	MCWBR	8.5	9.4	8.8	9.2	9.9	9.7	9.3	8.7	9.0	7.7	7.7	7.6		
	taub		0.321	0.344	0.373	0.408	0.413	0.413	0.421	0.414	0.386	0.359	0.338	0.317		
		taud	2.412	2.374	2.306	2.238	2.259	2.276	2.253	2.299	2.366	2.434	2.416	2.385		
(42) Clear-Sky Solar Irradiance	Ebn at Noon		220	246	259	263	267	268	263	258	252	237	210	200		
		Edh at Noon	19	26	33	40	41	40	41	37	31	24	19	17		
	RadAvg		233	445	800	1288	1557	1676	1596	1358	993	585	275	177		
(45) All-Sky Solar Radiation	RadStd		35	85	122	146	141	149	201	126	118	54	36	23		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (78 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page